

OCTOBER 2026 UPDATE
ANNA SOFIA MORO

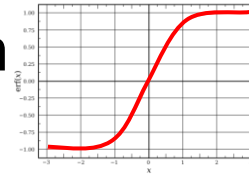
ABOUT MULTIPLE COULOMB SCATTERING

- Study the angular deflection distribution $\Delta\theta$ to measure MCS
- Mapping the local angular spread $\sigma(\Delta\theta)$ as a function of x, y

what to expect?

material transitions appear as sharp steps

- Model the transition with an Error function
- Compare $\sigma(\Delta\theta)$ with the theory:



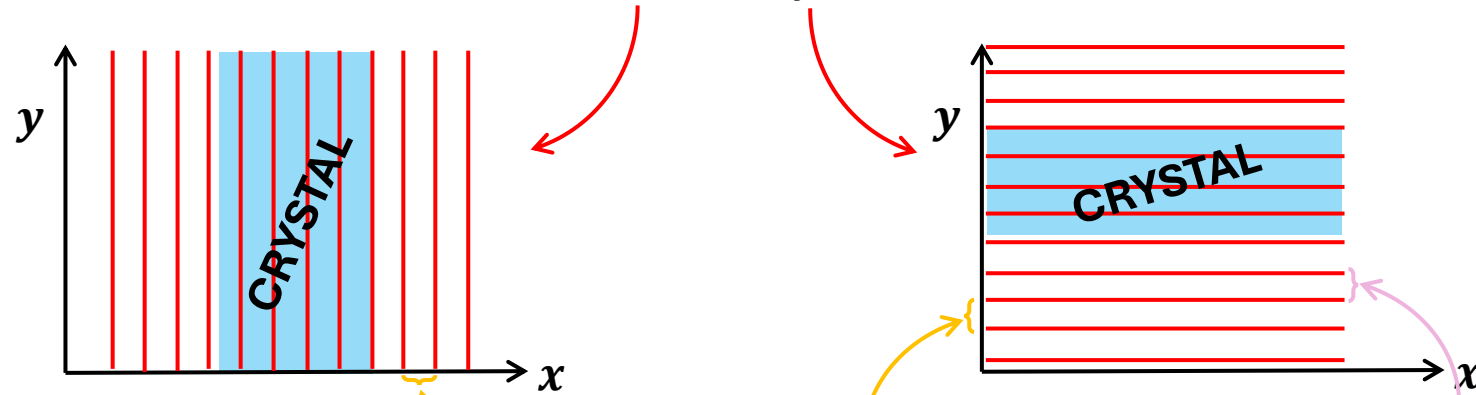
$$\langle\theta_{MCS}\rangle = \frac{13.6 \text{ MeV}}{\beta cp} z \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{xz^2}{X_0\beta^2} \right) \right]$$

inside: $\theta_{Si} = 66.5 \mu\text{rad}$

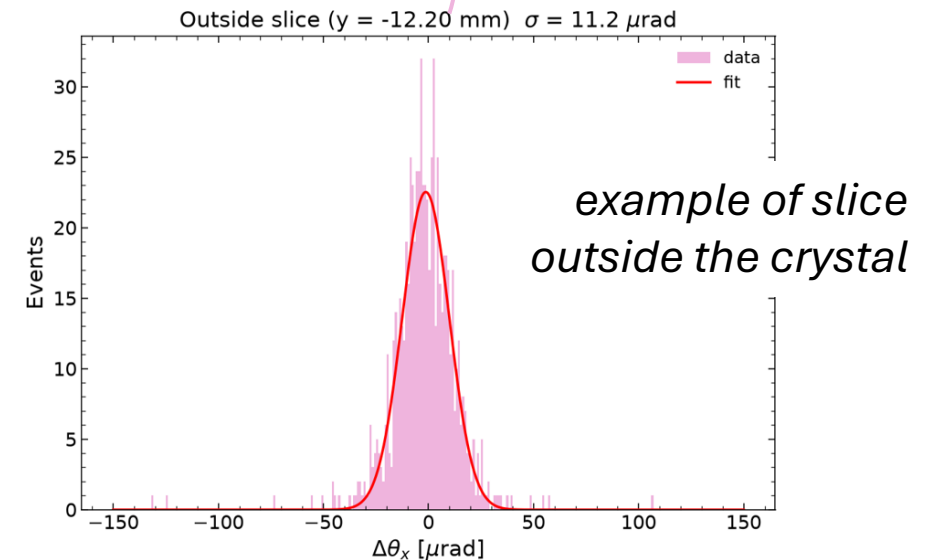
outside: $\theta_{air} = 0.8 \mu\text{rad}$

METHOD

- Divided the crystal into slices along x and y



- Width of the angular distribution σ for each slice
 - I always expect a distribution centered at zero
 - Gaussian fit within $\mu \pm 2.5\sigma$
 - The measured width also accounts for the tracker uncertainty of $12 \mu\text{rad}$.



METHOD

The measured width also accounts for the tracker uncertainty of $12 \mu\text{rad}$.

↓ how to take it into account?

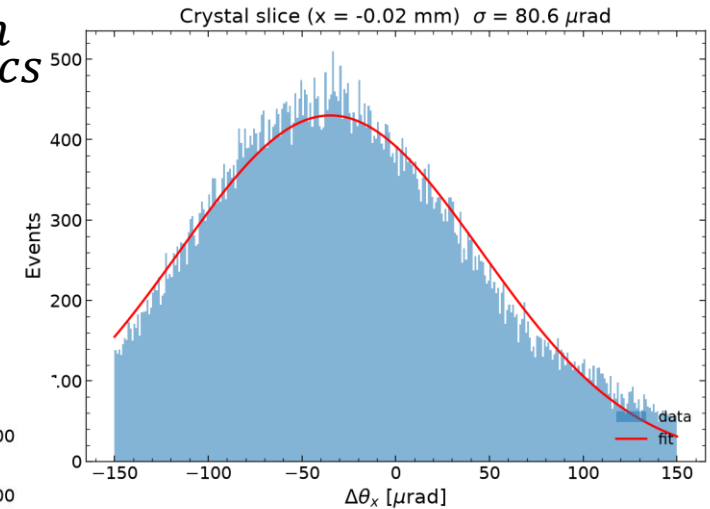
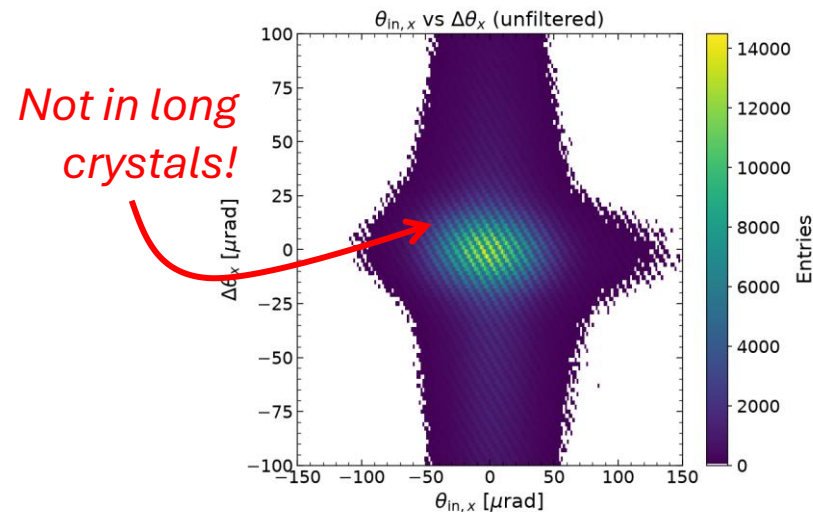
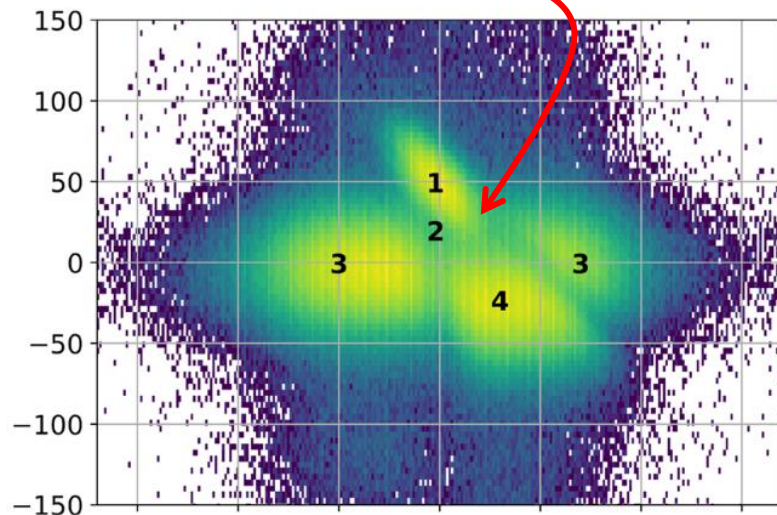
- We are measuring spreads of distributions → they are variances.
- Tracker contribution is independent → add in quadrature.

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2$$
$$\rho = 0 \text{ (independent): } \sigma^2 = \sigma_1^2 + \sigma_2^2$$

METHOD

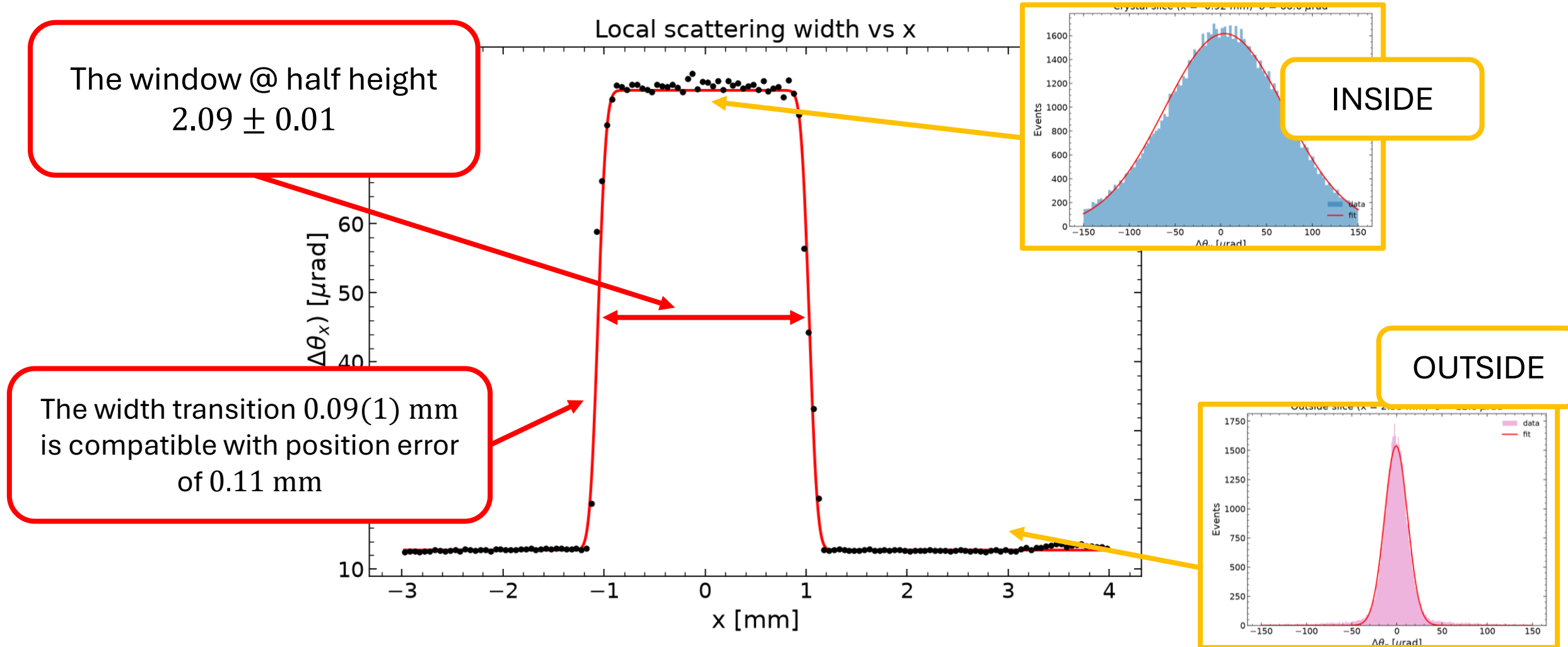
Problem: distribution not centered at zero and $\sigma_{meas} \gg \sigma_{mcs}^{th}$

- Unreliable results
- filter on the incoming angle?
 - In short crystals we are able to indentify and filter the regions



Solution: use only $\Delta\theta_y$ instead of $\Delta\theta_x$ to have pure MCS (isotropic)

RESULTS ALONG x



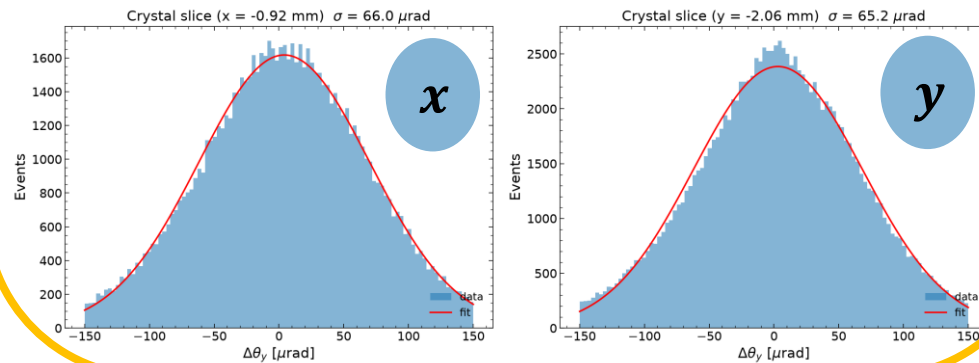
RESULTS

INSIDE

$$\theta_{MCS}^{th} = 66.5 \mu\text{rad}$$

$$\sigma_{meas} = \sqrt{\sigma_{mcs}^2 + \sigma_{tracker}^2}$$

$$\theta_{MCS}^{meas} = \sqrt{\sigma_{fit}^2 - \sigma_{tracker}^2} = 65 \mu\text{rad}$$

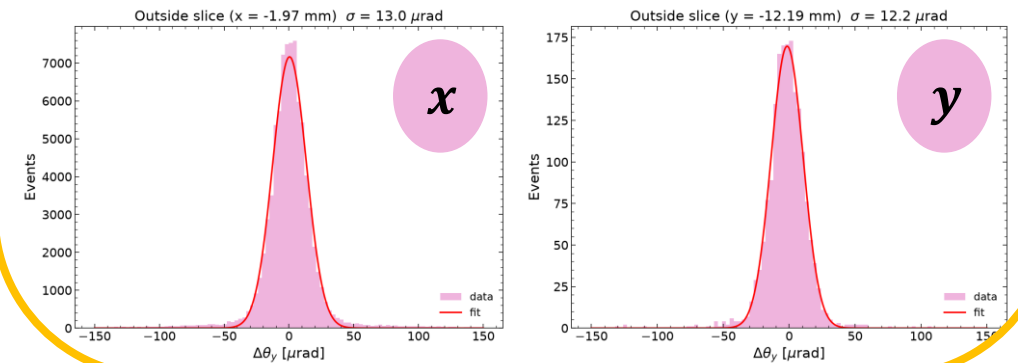


OUTSIDE

$$\theta_{MCS}^{th} = 0.8 \mu\text{rad}$$

$$\sigma_{meas} = \sqrt{\sigma_{air}^2 + \sigma_{track}^2}$$

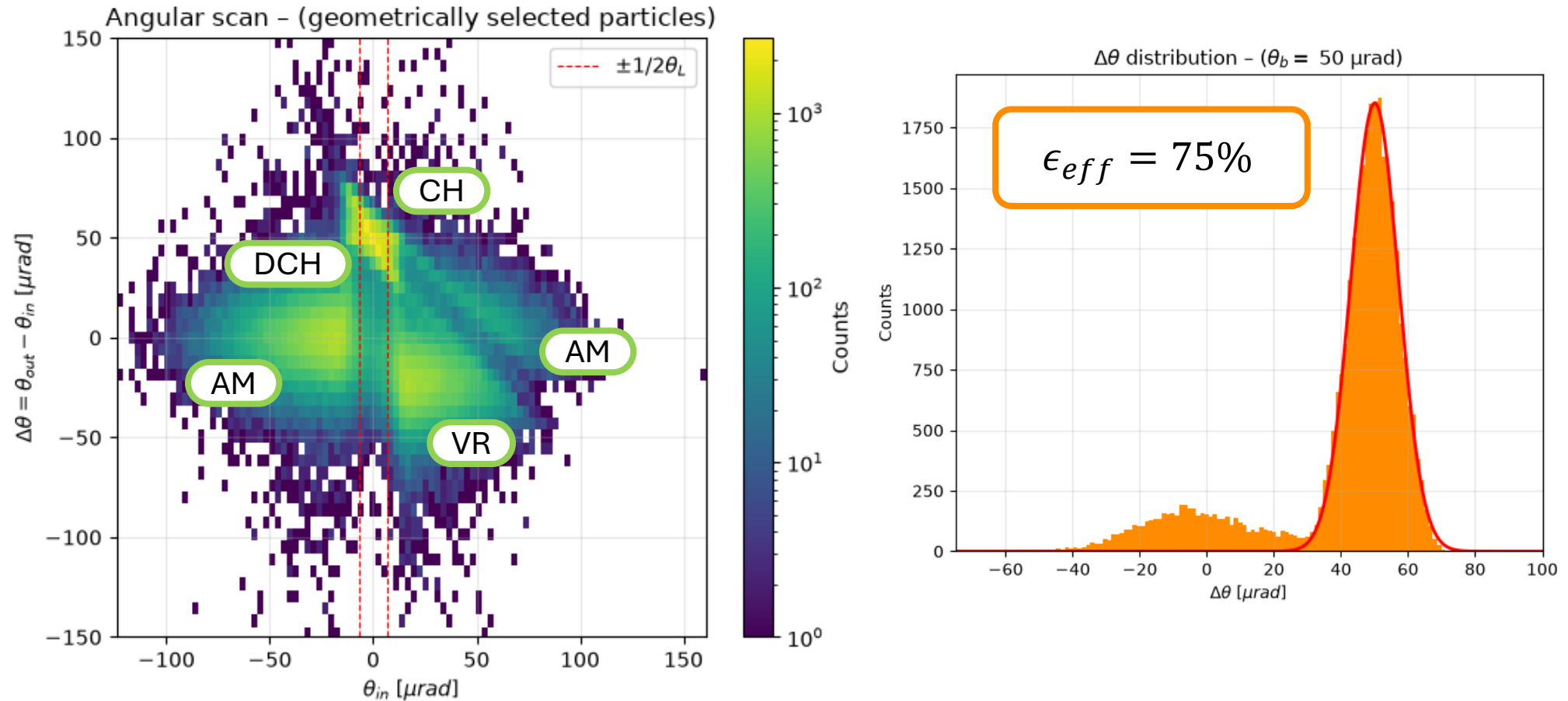
$$\theta_{MCS}^{meas} = \sqrt{\sigma_{fit}^2 - \sigma_{tracker}^2} = 1 \mu\text{rad}$$



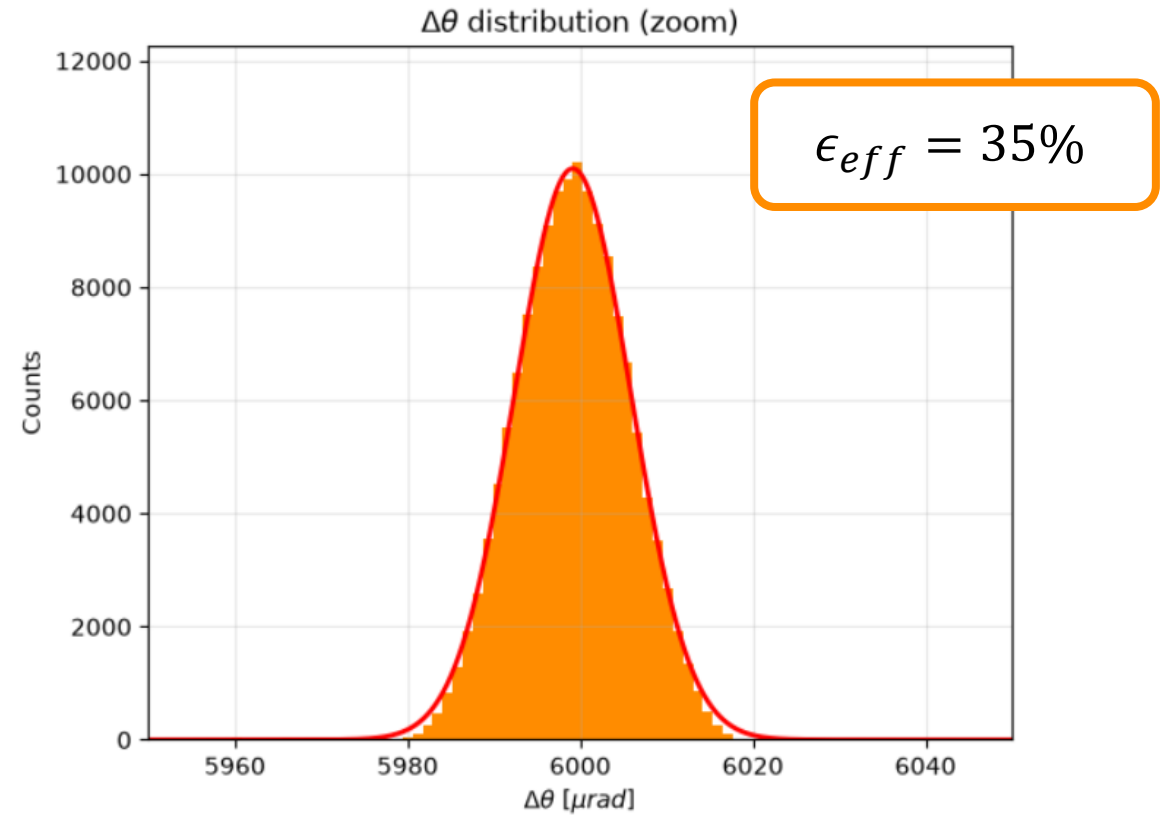
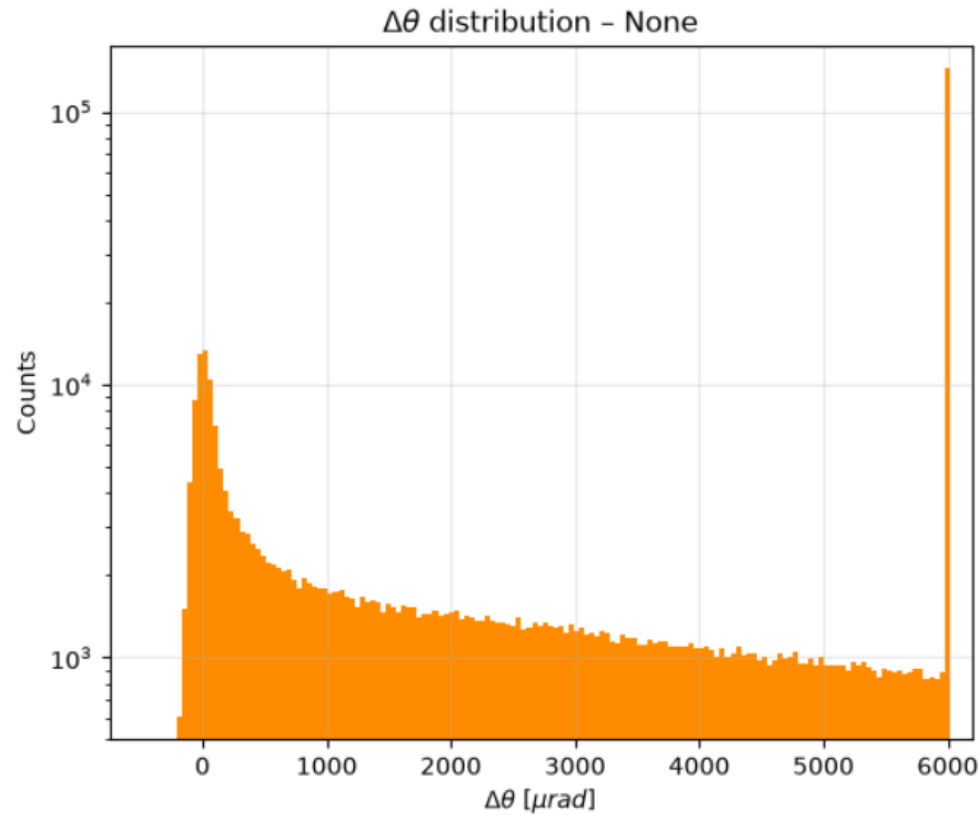
ABOUT THE SIMULATIONS

- General comparison:
 - Absorbed particles in Silicon due to inelastic nuclear reactions:
$$N \sim (1 - e^{-L/\lambda})$$
 - Particles entered in the crystal: $f = \operatorname{erf}\left(\frac{a}{\sigma_{\theta}\sqrt{2}}\right)$
- Short crystal (50 μrad):
 - should I simulate them with different seeds?
- Long crystal (6 mrad):
 - Use another routine? Xtrack?

SIMULATING THE SHORT CRYSTAL

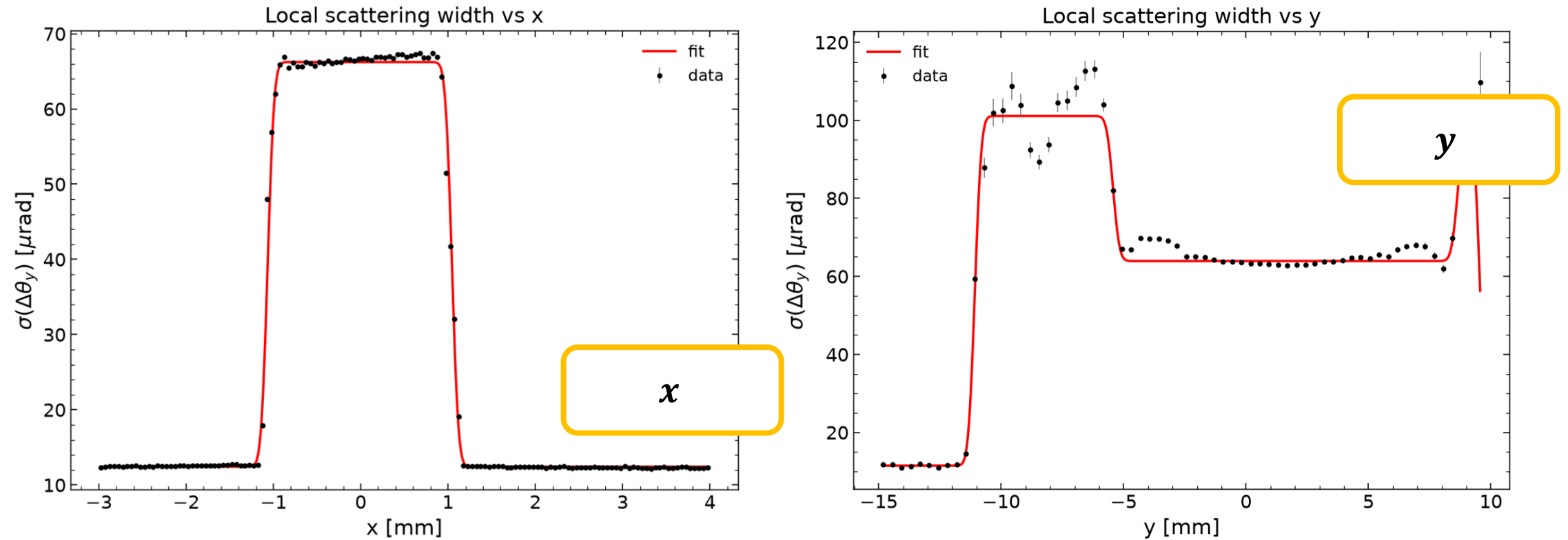


SIMULATING THE LONG CRYSTAL



BACKUP

RESULTS



RESULTS

